

CSE 717 — Information Security

Topic 4: Hash Functions & Hash Table Collision Resolution

Complete Past Paper Solutions: 2020–2024

Source: *LcMaterial#2_InfoSecuritySP26.pdf* (Hashing, pp.2–25)

Every Topic-4 question 2020–2024 included. Zero omissions.

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Quick Reference — Hashing Cheat Sheet

What hashing is for (Lc#2 pp.2–10)

- **Hash table:** stores key→value pairs (like labelled lockers — need the right key to access a slot).
- **Hashing is irreversible** (unlike encryption, which is reversible) — used for **security, integrity, authentication**, NOT for hiding data.
- **Applications:** (1) Password storage — store $\text{hash}(\text{password})$, compare hashes on login; (2) Data integrity — compare file hash before/after transfer; (3) Digital signatures — sign the hash with private key; (4) MAC/HMAC — verify message integrity + sender authenticity; (5) Blockchain — block linking, Proof-of-Work, tamper resistance.

Properties of a Good Hash Function (6, Lc#2 pp.11–13)

1. **Deterministic:** same input $\rightarrow$ always the same output hash.
2. **Quick Computation:** hash must be computable fast, or the system isn't efficient.
3. **Pre-Image Resistance:** given $H(A)$, infeasible to recover A .
4. **Small change $\rightarrow$ big hash change (avalanche):** a tiny change in input causes a huge change in the output hash.
5. **Collision Resistant:** infeasible to find $A \neq B$ with $H(A) = H(B)$.
6. **Puzzle Friendly:** hard to pick an input that produces a pre-defined output — inputs should be selectable from as wide a distribution as possible.

Collision-Resolution Methods (Open Addressing)

- **Linear Probing:** if slot $h(k) \bmod m$ is full, try $(h(k) + 1) \bmod m$, then $+2$, $+3$, $\dots$ — check the *next* slot sequentially.
- **Quadratic Probing:** if $h(k) \bmod m$ is full, try $(h(k) + 1^2) \bmod m$, then $(h(k) + 2^2) \bmod m$, then $(h(k) + 3^2) \bmod m$, $\dots$
- **Double Hashing:** use a 2nd hash function $h_2(k)$ for the step size — probe sequence is

$$h_{\text{new}} = (h_1(k) + i \cdot h_2(k)) \bmod m, \quad i = 0, 1, 2, \dots$$

(most collision-resistant of the three — step size varies per key, avoids clustering).

2024 — Section B Q7(c): Double Hashing Insertion (3 marks)

Question — 2024 Section B Q7(c)

Insert the keys 8, 47, 22, 44, 39, 32 (in this order) into a hash table of size $m = 13$ using **double hashing**, with $h_1(k) = k \bmod 13$ and $h_2(k) = 7 + (k \bmod 7)$. Show the resultant hash table. [3]

Answer — 2024 Section B Q7(c)

Step 1 — Compute $h_1(k)$ and $h_2(k)$ for each key:

k	$h_1(k) = k \bmod 13$	$h_2(k) = 7 + (k \bmod 7)$
8	8	$7 + 1 = 8$
47	8	$7 + 5 = 12$
22	9	$7 + 1 = 8$
44	5	$7 + 2 = 9$
39	0	$7 + 4 = 11$
32	6	$7 + 4 = 11$

Step 2 — Insert in order (probe sequence $h_{\text{new}} = (h_1(k) + i \cdot h_2(k)) \bmod 13$, $i = 0, 1, 2, \dots$):

- **8:** $h_1 = 8$, slot 8 empty → **place at 8.**
- **47:** $h_1 = 8$, slot 8 *occupied* (by 8). Try $i = 1$: $(8 + 12) \bmod 13 = 20 \bmod 13 = 7$, slot 7 empty → **place at 7.**
- **22:** $h_1 = 9$, slot 9 empty → **place at 9.**
- **44:** $h_1 = 5$, slot 5 empty → **place at 5.**
- **39:** $h_1 = 0$, slot 0 empty → **place at 0.**
- **32:** $h_1 = 6$, slot 6 empty → **place at 6.**

Final hash table ($m = 13$):

Slot	0	1	2	3	4	5	6	7	8	9	10	11	12
Key	39	–	–	–	–	44	32	47	8	22	–	–	–

Only key 47 collided (with 8 at slot 8) and needed 1 extra probe via h_2 .

2023 — Section B Q8(c): Linear Probing Insertion (3 marks) + Q5(b): Hash Function Properties (2.5 marks)

Question — 2023 Section B Q8(c)

Insert the keys 21, 8, 13, 44, 28, 33 (in this order) into a hash table of size $m = 13$ using **open addressing with linear probing**, with primary hash function $h(k) = (k + 2) \bmod 13$. Show the resultant hash table. [3]

Answer — 2023 Section B Q8(c)

Step 1 — Compute $h(k) = (k + 2) \bmod 13$ for each key:

k	$k + 2$	$h(k) = (k + 2) \bmod 13$
21	23	10
8	10	10
13	15	2
44	46	7
28	30	4
33	35	9

Step 2 — Insert in order (linear probe: if slot full, try next slot +1, wrap around mod 13):

- **21:** $h = 10$, slot 10 empty → **place at 10.**
- **8:** $h = 10$, slot 10 *occupied* (by 21). Probe slot 11 — empty → **place at 11.**
- **13:** $h = 2$, slot 2 empty → **place at 2.**
- **44:** $h = 7$, slot 7 empty → **place at 7.**
- **28:** $h = 4$, slot 4 empty → **place at 4.**
- **33:** $h = 9$, slot 9 empty → **place at 9.**

Final hash table ($m = 13$):

Slot	0	1	2	3	4	5	6	7	8	9	10	11	12
Key	—	—	13	—	28	—	—	44	—	33	21	8	—

Only key 8 collided (with 21 at slot 10) and needed 1 extra linear probe to slot 11.

Part Q5(b): Properties of a Good Hash Function

Question — 2023 Section B Q5(b)

Enumerate the properties of a good hash function.

[2.5]

Answer — 2023 Section B Q5(b)

1. **Deterministic:** same input always produces the same hash output.
2. **Quick Computation:** the hash must be computed efficiently/quickly for any input.
3. **Pre-Image Resistance:** given $H(A)$, it is infeasible to find A .
4. **Avalanche effect:** a small change in input causes a large, unpredictable change in the output hash.
5. **Collision Resistant:** infeasible to find two different inputs $A \neq B$ with $H(A) = H(B)$.
6. **Puzzle Friendly:** infeasible to choose an input that yields a pre-defined output — outputs appear uniformly distributed.

For 2.5 marks, naming + one-line definition of 5 of these 6 is sufficient — pick the first 5 if pressed for time.

Exam Strategy Summary

High-Yield Checklist for Hash Functions & Collision Resolution (2/5 years, 2023+2024)

1. **Both 2023 and 2024 ask a hash-table insertion trace** — 2023 used **linear probing** (open addressing, simple $h(k) = (k + c) \bmod m$), 2024 used **double hashing** ($h_1 + i \cdot h_2 \bmod m$). **Practice BOTH** on a fresh 6-key list with $m = 10$ – 13 before the exam.
2. **Mechanical drill:** for each key (in order) — compute primary hash, check if slot is free; if occupied, apply the probe sequence (linear: $+1, +2, \dots$; double hashing: $+i \cdot h_2(k)$) until an empty slot is found.
3. **Hash function properties** (2023 Q5b, 2.5 marks): memorize the 6-term list — Deterministic, Quick Computation, Pre-Image Resistance, Avalanche, Collision Resistant, Puzzle Friendly. Cheap, pure-recall marks.
4. **Quadratic probing** ($f(i) = i^2$) appears in Lc#2 with worked examples but NOT in 2020–2024 papers — know the formula as a backup, don't over-invest.
5. **Don't confuse the three methods:** linear probing = fixed step $+1$; quadratic probing = step i^2 ; double hashing = step $i \cdot h_2(k)$ (a second, key-dependent hash function).